Machine Learning Regression on Synthetic Gravitational-Wave Inspirals

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Motivation

- Gravitational-wave (GW) astronomy relies on recovering source parameters from noisy detector data
- Classical parameter estimation (e.g. LIGO's matched filtering + Bayesian inference) is **computationally expensive**
- **Goal:** Build an ML surrogate model that maps GW inspiral features to source parameters

What are Gravitational Waves?

Ripples in spacetime produced by accelerating masses. First predicted by Einstein's General Relativity (1916), first directly detected by LIGO (2015).

Key properties:

- Travel at the speed of light
- Stretch and squeeze space transversely
- Carry energy away from the source
- Strongest from compact, violent events: binary black holes, neutron star mergers

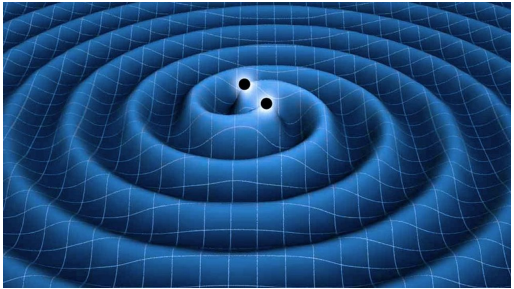


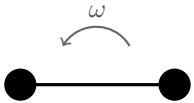
Figure: *[Binary Black Hole merger Simulation]*

Radiation and the Motion of Two Point Masses by Peters 1964

- Question: When two massive objects (like stars) orbit each other, does general relativity predict they will slowly spiral together? And if so, how fast?
- Peters calculated exactly how much energy and angular momentum are radiated per orbit.
- Using those energy-loss calculations, he derived equations describing how the orbit changes over time.
- In this section of the paper Peters treats the system of two point masses moving in elliptical orbits under their mutual gravitational attraction.
- Hulse and Taylor won the Nobel Prize in 1993 for this.

Quadrupole Approximation

- A **quadrupole moment** describes the asymmetric shape of a mass distribution
- This changing configuration **radiates gravitational waves** (Creighton 2026)



Rotating dumbbell

Analogy: the shape of a dumbbell changes as it rotates or evolves

Key Assumptions

- Weak-field regime
- Non-relativistic velocities ($v \ll c$)
- System size $\ll$ gravitational wavelength

Physical Parameters

System Parameters

- Masses: $m_1, m_2 \in [1.2, 50]\, M_\odot$
- Initial Separation: $r_0 \in [5 \times 10^6, 10^8]$ km
- Distance to observer: $d \in [10, 10^5]$ kpc

Physical Constants

- $G = 6.674 \times 10^{-11}\ \mathrm{m}^3\ \mathrm{kg}^{-1}\ \mathrm{s}^{-2}$
- $c = 3 \times 10^8$ m/s
- $M_\odot = 1.989 \times 10^{30}$ kg

- Circular orbits assumed ($e \approx 0$), consistent with compact binaries (Peters 1964)
- **1,000 synthetic samples** generated in total

Orbital Decay & Frequency

Orbital Decay (Peters 1964)

$$\frac{dr}{dt} = -\frac{64}{5}\frac{G^3 m_1 m_2 (m_1 + m_2)}{c^5 r^3}$$

- Tracks how separation shrinks with time
- Solved numerically via `scipy` RK45

Orbital Frequency (derived in the report)

$$f_{\text{orb}} = \frac{1}{2\pi}\sqrt{\frac{G(m_1 + m_2)}{r^3}}$$

- Increases as separation decreases
- Derived from $F_g = F_{\text{cen}}$

Strain Amplitude & Target Labels

GW Strain (Creighton 2026)

$$h(t) = \frac{4(G\mathcal{M})^{5/3}\,\omega_r^{2/3}}{c^4\,d}$$

- Amplitude of gravitational waves
- orbital angular frequency: $\omega_r = 2\pi f_{\text{orb}}$

Target Labels

Chirp mass:

$$\mathcal{M} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}$$

Merger time (derived in the report):

$$t_m = \frac{5}{256}\frac{c^5 r_0^4}{G^3 m_1 m_2 (m_1 + m_2)}$$

Dataset & Feature Engineering

20 Input Features

- noisy strain time-series: $h_{\text{noisy}}(t_1), \ldots, h_{\text{noisy}}(t_{17})$
- distance to observer: d
- initial separation: r_0
- reduced mass: $\mu = \frac{m_1 m_2}{m_1 + m_2}$

Preprocessing

- No missing values (synthetic data)
- **Log-transform** applied due to large SI-unit magnitudes
- Values converted back to linear space for Mean Absolute Error
- **75/25 train-test split** with fixed random seed

Why Random Forest?

- **Ensemble method:** combines many decision trees for better generalization (Kavlakoglu 2025)
- **Non-linearity:** strain $\rightarrow$ chirp mass / merger time mapping is highly non-linear
- **Scale-invariance:** features span vastly different physical scales and RF handles this naturally
- **Overfitting mitigation:** ensemble averaging reduces variance with only 1,000 samples
- Implementation: `scikit-learn` `RandomForestRegressor`

Experiment 1 — Full Feature Set

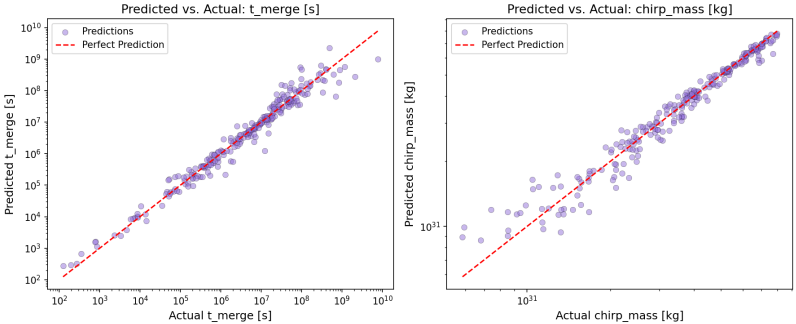


Figure 1: Predicted vs. Actual — full feature set

- Decent training fit but poor generalization
- Model **memories** r_0 **values**; struggles at extremes
- Large MAE reflects SI-unit scale

	Train	**Test**
MAE	4.1×10^{29}	1.1×10^{30}
R^2	0.87	0.59

Feature Importance — Full Feature Set

- r_0 **dominates** ($\sim$75%) for merger time prediction and μ **is second** ($\sim$25%) for chirp mass prediction
- Both appear directly in the *analytical formulae*:
  - $t_m \propto r_0^4$
  - $\mathcal{M}$ depends on μ
- Strain features contribute **effectively nothing**
- Model learned **algebra, not physics**

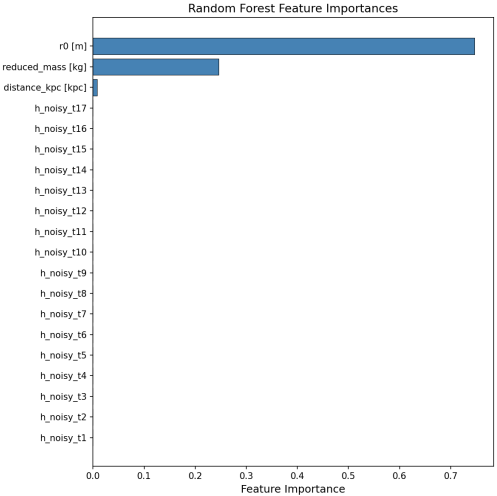


Figure 3b: Feature importances — full feature set

Experiment 2 — Strain-Only Features

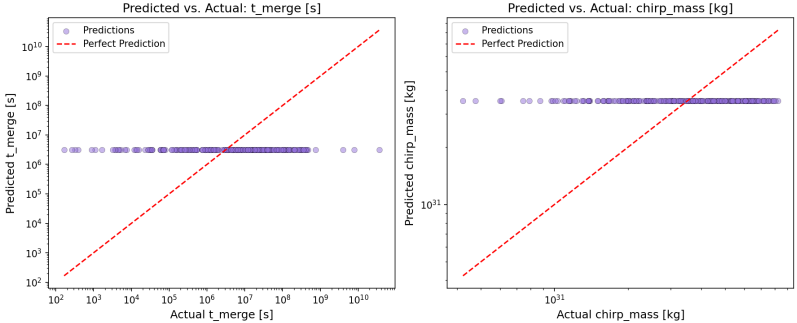


Figure 2: Predicted vs. Actual — strain-only features

- Removed r_0, d, μ to test whether strain alone contains learnable signal
- Increased strain samples from 17 to **20 time-stamps**
- Predictions collapse to a *flat horizontal line* indicating the dataset mean
- Classic signature of a model that **found no learnable signal**

Feature Importance — Strain-Only

- All feature importances are effectively **zero**
- Apparent negative values are a *plotting artifact* of axis scaling
- Model cannot extract any meaningful signal from 20 noisy strain snapshots
- Confirms: raw strain time-series is **insufficient** for this regressor

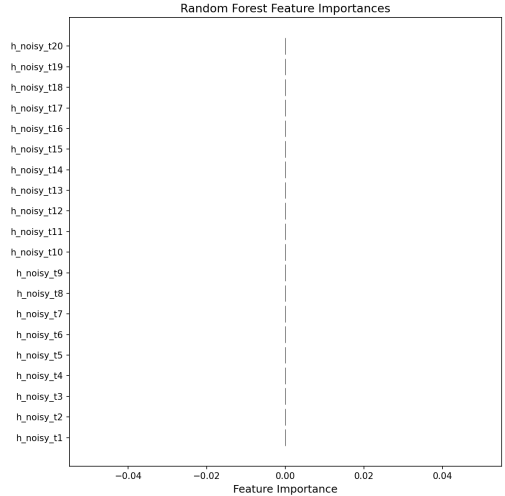


Figure 3a: Feature importances — strain-only

Conclusion

Key Findings

- A Random Forest with **full features** ($R^2 = 0.87$ train) was exploiting algebraic shortcuts, not learning from strain
- Restricting to **strain only** caused complete model failure ($R^2 \approx 0$)
- Real GW parameter estimation uses **matched filtering** + Bayesian inference (LALInference (Veitch et al. 2015), Bilby (Ashton et al. 2019)) — far beyond a simple regressor

Lessons Learned

- Including analytically-derived features *leaks* the answer to the model
- 17–20 noisy time samples are **too sparse** to capture waveform morphology
- Feature importance is a vital diagnostic tool

Different Approach

- Generate a **denser strain time-series** (hundreds of time steps spanning the full inspiral)
- **Normalise strain by distance** to remove d as an implicit shortcut
- Explore **other model architectures** that can extract waveform morphology

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